

2.4" TFT Display Module with Touch (240x320)

👉 Introduction

The **2.4-inch SPI TFT Display Module with ST7789V** is a high-resolution, full-color LCD designed for embedded applications. Featuring a 240 × 320-pixel LCD screen, it offers wide viewing angles and vibrant color reproduction. The ST7789V driver IC enables efficient communication via SPI, making it compatible with STM32, ESP32, Raspberry Pi, and Arduino. With low power consumption, a high refresh rate, and an optional touchscreen, this module is ideal for graphical user interfaces (GUIs), real-time data visualization, and portable embedded systems.

👉 Feature

1. **Display Type:** 2.4-inch TFT LCD
2. **Resolution:** 240 × 320 pixels
3. **Driver IC:** ST7789V
4. **Interface:** SPI
5. **Data Format:** RGB565
6. **Touch Support:** Resistive (XPT2046)

👉 Arduino Hardware Connection

TFT Display	Arduino Pin
VCC	5V
GND	GND
RESET	8
SCK	13
DC	9
CS	10
SDI(MOSI)	11
LED	5V
SDO(MISO)	12

XPT2046 Pin	Arduino Pin
T_CLK	13
T_CS	4
T_DIN	11
T_DO	12
T_IRQ	Not Required

👉 Arduino Demo Code

```
#include <SPI.h>
#include <Adafruit_GFX.h>
#include <Adafruit_ST7789.h>
#include <XPT2046_Touchscreen.h>

// Pin Definitions
#define TFT_MOSI 11 // Data out
#define TFT_SCLK 13 // Clock out
#define TFT_CS 10 // Chip select
#define TFT_RST 8 // Reset pin
#define TFT_DC 9 // Data/Command pin
#define TOUCH_CS 4

// Initialize Display and Touch Objects
Adafruit_ST7789 tft = Adafruit_ST7789(TFT_CS, TFT_DC, TFT_MOSI, TFT_SCLK,
TFT_RST);

XPT2046_Touchscreen ts(TOUCH_CS);

void setup() {
  Serial.begin(115200);

  // Initialize Display
  tft.init(240, 320); // Set display size
  tft.setRotation(3); // Adjust rotation (0-3)

  // Initialize Touchscreen
  ts.begin();
  ts.setRotation(3); // Adjust for correct touch mapping

  // Draw UI with Buttons
  drawUI();
}

void loop() {
  if (ts.touched()) {
    TS_Point p = ts.getPoint(); // Get touch point

    // Convert raw touch coordinates to screen pixels
    int touchX = map(p.x, 300, 3800, 0, 240);
    int touchY = map(p.y, 200, 3800, 0, 320);

    Serial.print("X: "); Serial.print(touchX);
    Serial.print(" Y: "); Serial.println(touchY);
  }
}
```

```

        // Check if a button is pressed
        checkButtons(touchX, touchY);

        delay(200); // Debounce delay
    }
}

// Function to Draw UI with Buttons
void drawUI() {
    tft.fillScreen(ST77XX_BLACK);
    tft.setTextColor(ST77XX_WHITE);
    tft.setTextSize(2);

    // Draw Buttons
    tft.fillRect(50, 100, 140, 50, ST77XX_BLUE); // Button 1
    tft.fillRect(50, 180, 140, 50, ST77XX_RED); // Button 2

    // Label Buttons
    tft.setCursor(75, 120);
    tft.print("BUTTON 1");

    tft.setCursor(75, 200);
    tft.print("BUTTON 2");
}

// Function to Check Button Press
void checkButtons(int x, int y) {
    if (x > 50 && x < 190 && y > 100 && y < 150) { // Button 1
        tft.fillRect(50, 100, 140, 50, ST77XX_GREEN);
        tft.setCursor(75, 120);
        tft.setTextColor(ST77XX_BLACK);
        tft.print("BUTTON 1");
        delay(300);
        drawUI();
    }
    else if (x > 50 && x < 190 && y > 180 && y < 230) { // Button 2
        tft.fillRect(50, 180, 140, 50, ST77XX_YELLOW);
        tft.setCursor(75, 200);
        tft.setTextColor(ST77XX_BLACK);
        tft.print("BUTTON 2");
        delay(300);
        drawUI();
    }
}

```